

P1072R4

basic_string::resize_default_init

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This version:

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Abstract

Allow access to default-initialized elements of basic_string.

Table of Contents

- 1 Motivation**
- 2 Proposal**
- 3 Implementation**
- 4 Examples**
 - 4.1 Stamping a Pattern
 - 4.2 Interacting with C
- 5 Design Considerations**
 - 5.1 Method vs. Alternatives

5.2 Allocator Interaction

5.3 Bikeshedding

6 Alternatives Considered

6.1 Tag Type

6.2 Non-Lambda Approach

6.3 Standalone Type: `storage_buffer`

6.4 Externally-Allocated Buffer Injection

7 Related Work

7.1 Google

7.2 MongoDB

7.3 VMware

7.4 Discussion on std-proposals

7.5 DynamicBuffer

7.6 P1020R1

7.7 Boost

7.8 Thrust

8 Wording

8.1 `[basic.string]`

8.2 `[container.requirements.general]`

9 Questions for LEWG

10 History

10.1 R3 → R4

10.2 R2 → R3

10.3 R1 → R2

10.4 R0 → R1

11 Acknowledgments

References

Informative References

§ 1. Motivation

Performance sensitive code is impacted by the cost of initializing and manipulating strings. When streaming data into a `basic_string`, a programmer is faced with an unhappy choice:

- **Pay for extra initialization** — `resize`, which zero initializes, followed by copy.
- **Pay for extra copies** — Populate a temporary buffer, copy it to the string.
- **Pay for extra "bookkeeping"** — `reserve` followed by small appends, each of which checks capacity and null terminates.

The effects of these unhappy choices can all be measured at scale, yet C++'s hallmark is to leave no room between the language (or in this case the library) and another language.

LEWGI polled on this at the [\[SAN\]](#) Meeting:

We should promise more committee time to [\[P1072R1\]](#), knowing that our time is scarce and this will leave less time for other work?

Unanimous consent

LEWG at the [\[SAN\]](#) Meeting:

We want to solve this problem

SF	F	N	A	SA
17	5	0	0	0

Willing to solve this for string without solving it for vector

SF	F	N	A	SA
6	9	2	1	0

§ 2. Proposal

This proposal addresses the problem by adding `basic_string::resize_default_init`:

```
template  
void resize_default_init(size_type n, Operation op);
```

8. *Throws:* `out_of_range` if `op(data(), n) > n`.

9. *Effects:* Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` default-initialized elements.

Invokes `erase(op(data(), n))`.

10. *Remarks:* Let `m = op(data(), n)`. `op` shall replace (`expr.ass`) values in the range `[data(), data()+m)`.

In order to enable `resize_default_init`, this proposal makes it implementation-defined whether `basic_string` uses `allocator_traits::construct` and `allocator_traits::destroy` to construct and destroy the "char-like objects" that it controls. See [§ 5.2 Allocator Interaction](#) and [§ 8 Wording](#) below for more details.

§ 3. Implementation

`libc++` includes a private implementation of this proposal (based on an earlier revision) and uses it to avoid a dynamic allocation in `std::filesystem [LIBC++]`.

§ 4. Examples

§ 4.1. Stamping a Pattern

Consider writing a pattern several times into a string:

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    ret.reserve(pattern.size() * count);
    for (size_t i = 0; i < count; i++) {
        // SUB-OPTIMAL:
        // * Writes 'count' nulls
        // * Updates size and checks for potential resize 'count' times
        ret.append(pattern);
    }

    return ret;
}

```

Alternatively, we could adjust the output string's size to its final size, avoiding the bookkeeping in `append` at the cost of extra initialization:

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    const auto step = pattern.size();
    // SUB-OPTIMAL: We memset step*count bytes only to overwrite them.
    ret.resize(step * count);
    for (size_t i = 0; i < count; i++) {
        // GOOD: No bookkeeping
        memcpy(ret.data() + i * step, pattern.data(), step);
    }

    return ret;
}

```

With this proposal:

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    const auto step = pattern.size();
    // GOOD: No initialization
    ret.resize_default_init(step * count, [&](char* buf, size_t n) {
        for (size_t i = 0; i < count; i++) {
            // GOOD: No bookkeeping
            memcpy(buf + i * step, pattern.data(), step);
        }
    });

    return ret;
}

```

§ 4.2. Interacting with C

Consider wrapping a C API while working in terms of C++'s `basic_string` vocabulary. We *anticipate* over-allocating, as computation of the *length* of the data is done simultaneously with the computation of the *contents*.

```
extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size)
    size_t compressBound(size_t bound);
}

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());

    // SUB-OPTIMAL: Extra initialization
    compressed.resize(size);
    int ret = compress(&*compressed.begin(), &size, input.data(), input.size());
    if (ret != OK) {
        throw ...some error...
    }

    // Set actual size.
    compressed.erase(size);
    return compressed;
}
```

With this proposal:

```

extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size)
    size_t compressBound(size_t bound);
}

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());

    // GOOD: No initialization
    compressed.resize_default_init(size, [&](char* buf, size_t n) {
        int ret = compress(buf, &size, input.data(), input.size());
        if (ret != OK) {
            throw ...some error...
        }
        return n;
    });
    return compressed;
}

```

§ 5. Design Considerations

§ 5.1. Method vs. Alternatives

`resize_default_init` exposes users to UB if they read indeterminate ([**basic.indet**] (6.6.4)) values in the string buffer passed to `op`. Despite this foot-gun, `resize_default_init` is appealing because:

- Uninitialized buffers are not uncommon in C++. See for example `new T[20]` or the recently adopted `make_unique_default_init`.
- Dynamic analyzers like Memory Sanitizer [\[MSan\]](#) and Valgrind [\[Valgrind\]](#) can catch uninitialized reads.

During the [\[SAN\]](#) Meeting LEWG expressed a preference for implementing this functionality as a new method on `basic_string` (as proposed in [\[P1072R0\]](#)) rather than a standalone "storage buffer" type (one option in [\[P1072R1\]](#)):

Method on string vs storage_buffer type:

Strong Method	Method	Neutral	Type	Strong Type
9	2	3	2	2

During the [KOA] Meeting, LEWG expressed a preference for not weakening `basic_string`'s invariants after `resize_default_init` returns. Under this revision:

- A default-initialized buffer is *only* accessible to `op`.
- Under penalty of undefined behavior, `op` must assign (replace) the values in resulting buffer. The default initialized portion (`[data()+m, data()+n)`) above is erased and inaccessible to the program without further violation of class invariants.
- This proposal follows the "lambda" option discussed in [SAN].

Several other alternatives were discussed at [SAN], [KOA], and on the reflector. Please see [§ 6 Alternatives Considered](#) below for more details.

§ 5.2. Allocator Interaction

Unlocking the desired optimizations requires *some* change to `basic_string`'s interaction with allocators. This proposal does what we think is the simplest possible change: remove the requirement that `basic_string` call `allocator_traits::construct` or `allocator_traits::destroy`.

This restriction should be acceptable because `basic_string` is defined to only hold "non-array trivial standard-layout" types. [strings.general] p1:

This Clause describes components for manipulating sequences of any non-array trivial standard-layout (6.7) type. Such types are called *char-like types*, and objects of *char-like types* are called *char-like objects* or simply *characters*.

Removing calls to `construct` and `destroy` is compatible with `pmr::basic_string` as long as `uses_allocator_v<charT>` is `false`, which should be the case in practice.

Along the way, this proposal clarifies ambiguous language in [string.require] and [container.requirements.general] by:

- Stating explicitly that `basic_string` is allocator-aware.
- Introducing the term "allocator-aware" earlier in [container.requirements.general].

- Not attempting to mention other "components" in [`container.requirements.general`]. The other allocator-aware containers (just `basic_string` and `regex::match_results`?) can independently state that they are "allocator-aware" in their own clauses.

libstdc++ and msvc allow strings of non-trivial type. That might force those libraries to continue to support `construct` and `destroy` as "extensions". On the other hand, libc++ disallows non-trivial `charT`s, so any such extensions are non-portable. See godbolt.org.

§ 5.3. Bikeshedding

What do we want to call this method?

- `resize_default_init` (as proposed). This is consistent with `make_unique_default_init` [P1020R1], adopted in San Diego. Also, specifying default-initialization throws a bone to libraries that have historically allowed non-trivial `charT` types in `basic_string`.
- `resize_uninitialized` (as proposed in R0). "Uninitialized" is different from "default-initialized", which is what we want to specify. Also, `uninitialized` is already used elsewhere in the Library to mean "not constructed at all", as in `uninitialized_copy`; so this would be an overloading of the term.

§ 6. Alternatives Considered

§ 6.1. Tag Type

At the [SAN] Meeting, LEWG showed support for a tag argument type:

Approval (vote for as many as you find acceptable)

13	Go back to <code>resize_uninitialized</code>
15	Do tag type (<code>default_initialize</code>) for c'tor / <code>resize()</code>
12	Continue with <code>storage_buffer</code> (as per R2 of this paper)
7	Crack open with a lambda
7	RAII separate type

For example:

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    const auto step = pattern.size();

    // GOOD: No initialization
    std::string ret(step * count, std::string::default_init);
    for (size_t i = 0; i < count; i++) {
        // GOOD: No bookkeeping
        memcpy(ret.data() + i * step, pattern.data(), step);
    }

    return ret;
}

```

Benefits:

- A constructor that takes a tag type simplifies some use cases, like the example above.
- Matches existing practice in Boost. See [§ 7.7 Boost](#).

Drawbacks:

- Feature creep / complexity — A tag type invites generalizing for all allocator-aware container types. We agree that this is desirable and even has implementation experience in Boost. However default initialization is not enough. There is also "implicit construction" (see [\[P1010R1\]](#), [\[P0593R2\]](#)) and "relocation" (see [\[P1144R0\]](#), [\[P1029R1\]](#)). Neither of these features are yet in the language. It is too early to generalize. Note that the second poll quoted in [§ 1 Motivation](#) shows support for solving this problem for `[basic_]string` but not `vector`.
- In reflector discussion of `make_unique_default_init [Zero]`, there was a preference for avoiding tag types. The standard library has `copy_backward`, not `copy` with a tag, and `count_if`, rather than `count` with a predicate.

Conclusion:

LEWG should explore tags for allocator-aware containers, but that work should not block near-term enablement of efficient `[std::]string` builders.

§ 6.2. Non-Lambda Approach

In [\[P1072R0\]](#) and [\[P1072R3\]](#) of this proposal, the authors considered a method `resize_default_init` / `resize_uninitialized` which left a default-initialized buffer

accessible to users of the `basic_string` instance after the method returned. This method was rejected in [KOA], due to the weakened class invariants.

For illustration:

```
extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size)
    size_t compressBound(size_t bound);
}

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());

    // GOOD: No initialization
    compressed.resize_default_init(size);
    int ret = compress(&*compressed.begin(), &size, input.data(), input.size());
    if (ret != OK) {
        throw ...some error...
    }

    // Suppose size is the value of size before the call to compress and size'
    // is the value of size after the call to compress.
    //
    // If size' < size, then:
    //   std::cout << compressed[size' + 1]
    // ...triggers a read of uninitialized data.

    // Set actual size.
    compressed.erase(size);
    return compressed;
}
```

§ 6.3. Standalone Type: `storage_buffer`

In [P1072R1], we considered `storage_buffer`, a standalone type providing a `prepare` method (similar to the `resize_uninitialized` method proposed here) and `commit` (to promise, under penalty of UB, that the buffer had been initialized).

At the [\[SAN\]](#) Meeting, this approach received support from LEWGI in light of the [\[post-Rapperswil\]](#) email review indicating support for a distinct type. This approach was rejected by the larger LEWG room in San Diego Meeting Diego.

The proposed type would be move-only.

```
std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::storage_buffer<char> tmp;

    const auto step = pattern.size();
    // GOOD: No initialization
    tmp.prepare(step * count + 1);
    for (size_t i = 0; i < count; i++) {
        // GOOD: No bookkeeping
        memcpy(tmp.data() + i * step, pattern.data(), step);
    }

    tmp.commit(step * count);
    return std::string(std::move(tmp));
}
```

For purposes of the container API, `size()` corresponds to the *committed* portion of the buffer. This leads to more consistency when working with (and explicitly copying to) other containers via iterators, for example:

```
std::storage_buffer<char> buf;
buf.prepare(100);
*fill in data*
buf.commit(50);

std::string a(buf.begin(), buf.end());
std::string b(std::move(buf));

assert(a == b);
```

Benefits:

- By being move-only, we would not have the risk of copying types with trap representations (thereby triggering UB).

- Uninitialized data is only accessible from the `storage_buffer` type. For an API working with `basic_string`, no invariants are weakened. Crossing an API boundary with a `storage_buffer` is much more obvious than a "`basic_string` with possibly uninitialized data." Uninitialized bytes (the promise made by `commit`) never escape into the valid range of `basic_string`.

Drawbacks:

- `storage_buffer` requires introducing an extra type to the standard library, even though its novel functionality (from `string` and `vector`) is limited to the initialization abilities.
- `basic_string` is often implemented with a short-string optimization (SSO) and an extra type would need to implement that (likely by additional checks when moving to/from the `storage_buffer`) that are often unneeded.

§ 6.4. Externally-Allocated Buffer Injection

In [P1072R1], we considered that `basic_string` could "adopt" an externally `allocator::allocate`'d buffer. At the [SAN] Meeting, we concluded that this was:

- **Not critical**
- **Not constrained in the future**
- **Overly constraining to implementers.** Allowing users to provide their own buffers runs into the "offset problem". Consider an implementation that stores its `size` and `capacity` inline with its data, so `sizeof(container) == sizeof(void*)`.

```
class container {
    struct Rep {
        size_t size;
        size_t capacity;
    };

    Rep* rep_;
};
```

If using a `Rep`-style implementation, the mismatch in offsets requires an O(N) move to shift the contents into place and trigger a possible reallocation.

- **Introducing new pitfalls.** It would be easy to mix `new[]` and `allocator::allocate` inadvertently.

§ 7. Related Work

§ 7.1. Google

Google has a local extension to `basic_string` called `resize_uninitialized` which is wrapped as `STLStringResizeUninitialized`.

- [\[Abseil\]](#) uses this to avoid bookkeeping overheads in `StrAppend` and `StrCat`.
- [\[Snappy\]](#)
 - In [decompression](#), the final size of the output buffer is known before the contents are ready.
 - During [compression](#), an upperbound on the final compressed size is known, allowing data to be efficiently added to the output buffer (eliding `append`'s checks) and the string to be shrunk to its final, correct size.
- [\[Protobuf\]](#) avoids extraneous copies or initialization when the size is known before data is available (especially during parsing or serialization).

§ 7.2. MongoDB

MongoDB has a string builder that could have been implemented in terms of `basic_string` as a return value. However, as explained by Mathias Stearn, zero initialization was measured and was too costly. Instead a custom string builder type is used:

E.g.: <https://github.com/mongodb/mongo/blob/master/src/mongo/db/fts/unicode/string.h>

```

/**
 * Strips diacritics and case-folds the utf8 input string, as needed to sup
 * options.
 *
 * The options field specifies what operations to *skip*, so kCaseSensitive
 * means to skip case folding and kDiacriticSensitive means to skip diacrit
 * stripping. If both flags are specified, the input utf8 StringData is retu
 * directly without any processing or copying.
 *
 * If processing is performed, the returned StringData will be placed in
 * buffer.  buffer's contents (if any) will be replaced. Since we may retur
 * the input unmodified the returned StringData's lifetime is the shorter o
 * the input utf8 and the next modification to buffer. The input utf8 must
 * point into buffer.
 */
static StringData caseFoldAndStripDiacritics(StackBufBuilder* buffer,
                                             StringData utf8,
                                             SubstrMatchOptions options,
                                             CaseFoldMode mode);

```

(Comments re-wrapped.)

§ 7.3. VMware

VMware has an internal string builder implementation that avoids `std::string` due, in part, to `reserve`'s zero-writing behavior. This is similar in spirit to the MongoDB example above.

§ 7.4. Discussion on std-proposals

This topic was discussed in 2013 on std-proposals in a thread titled "Add `basic_string::resize_uninitialized` (or a similar mechanism)":

<https://groups.google.com/a/isocpp.org/forum/#!topic/std-proposals/XIO4KbBTxl0>

§ 7.5. DynamicBuffer

The [\[N4734\]](#) (the Networking TS) has *dynamic buffer* types.

§ 7.6. P1020R1

See also [\[P1020R1\]](#) "Smart pointer creation functions for default initialization". Adopted in San Diego.

§ 7.7. Boost

Boost provides a related optimization for vector-like containers, introduced in [\[SVN r85964\]](#) by Ion Gaztañaga.

E.g.: [boost/container/vector.hpp](#):

```
///! <b>Effects</b>: Constructs a vector that will use a copy of allocator a
///!   and inserts n default initialized values.
///!
///! <b>Throws</b>: If allocator_type's allocation
///!   throws or T's default initialization throws.
///!
///! <b>Complexity</b>: Linear to n.
///!
///! <b>Note</b>: Non-standard extension
vector(size_type n, default_init_t);
vector(size_type n, default_init_t, const allocator_type &a)
...
void resize(size_type new_size, default_init_t);
...
```

These optimizations are also supported in Boost Container's `small_vector`, `static_vector`, `deque`, `stable_vector`, and `string`.

§ 7.8. Thrust

The Thrust library has "a RAII-type `thrust::detail::temporary_array` which has a vector-like interface and a constructor with a tag parameter that indicates its elements should not be initialized." - [Bryce Adelstein Lelbach].

E.g. [thrust/thrust/detail/temporary_array.inl](#):

```

template<typename T, typename TemporaryArray, typename Size>
__host__ __device__
typename thrust::detail::disable_if<
    avoid_initialization<T>::value
>::type
    construct_values(TemporaryArray &a,
                    Size n)
{
    a.default_construct_n(a.begin(), n);
} // end construct_values()

```

§ 8. Wording

Wording is relative to [\[N4791\]](#).

Motivation for some of these edits can be found in [§ 5.2 Allocator Interaction](#).

§ 8.1. [\[basic.string\]](#)

In [\[basic.string\]](#) [20.3.2], in the synopsis, add [resize_default_init](#):

```

...
    // 20.3.2.4, capacity
    size_type size() const noexcept;
    size_type length() const noexcept;
    size_type max_size() const noexcept;
    void resize(size_type n, charT c);
    void resize(size_type n);
    template
    void resize_default_init(size_type n, Operation op);
    size_type capacity() const noexcept;
    void reserve(size_type res_arg);
    void shrink_to_fit();
    void clear() noexcept;
    [[nodiscard]] bool empty() const noexcept;

```

In [\[string.require\]](#) [20.3.2.1] clarify that [basic_string](#) is an allocator-aware container, and then add an exception for [construct](#) and [destroy](#).

3. In every specialization `basic_string<charT, traits, Allocator>`, the type `allocator_traits<Allocator>::value_type` shall name the same type as `charT`. ~~Every object of type `basic_string<charT, traits, Allocator>` uses an object of type `Allocator` to allocate and free storage for the contained `charT` objects as needed. The `Allocator` object used is obtained as described in [container.requirements.general]. In every specialization `basic_string<charT, traits, Allocator>`, and the type `traits`~~ shall satisfy the character traits requirements ([char.traits]). [Note: The program is ill-formed if `traits::char_type` is not the same type as `charT`. — end note]
4. `basic_string` is an allocator-aware container as described in [container.requirements.general], except that it is implementation-defined whether `allocator_traits::construct` and `allocator_traits::destroy` are used to construct and destroy the contained `charT` objects.
5. References, pointers, and iterators referring to the elements of a `basic_string` sequence may be invalidated by the following uses of that `basic_string` object:
...

In [string.capacity] [20.3.2.4]:

```
void resize(size_type n, charT c);
```

6. *Effects*: Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` copies of `c`.

```
void resize(size_type n);
```

7. *Effects*: Equivalent to `resize(n, charT())`.

```
void resize_default_init(size_type n);
```

8. *Effects*: Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` default-initialized elements.

9. *Remarks*: Let `m = op(data(), n)`. `op` shall replace (`[expr.ass]`) the values stored in the character array `[data(), data()+m)`.

```
size_type capacity() const noexcept;
```

```
...
```

§ 8.2. [container.requirements.general]

In [container.requirements.general] [21.2.1] clarify the ambiguous "components affected by this subclause" terminology in p3. Just say "allocator-aware containers".

3. All of the containers defined in this Clause except `array` are allocator-aware containers. ~~For the components affected by this subclause that declare an `allocator_type` Objects objects~~ stored in ~~these components~~ allocator-aware containers, unless otherwise specified, shall be constructed using the function `allocator_traits<allocator_type>::rebind_traits<U>::construct` and destroyed using the function `allocator_traits<allocator_type>::rebind_traits<U>::destroy` (19.10.9.2).

We can then simplify p15:

15. ~~All of the containers defined in this Clause and in 20.3.2 except `array` meet the additional requirements of an allocator-aware container, as~~ Allocator-aware containers meet the additional requirements described in Table 67.

§ 9. Questions for LEWG

1. Is LEWG satisfied with this approach?

§ 10. History

§ 10.1. R3 → R4

- Applied feedback from the [\[KOA\]](#) meeting.
- Moved to using a callback routine, rather than leaving invariants weakened from the time `resize_default_init` returned until values in the buffer were assigned.

§ 10.2. R2 → R3

- Applied Jonathan Wakely's editorial comments on [§ 8 Wording](#).
- Rebased on [\[N4791\]](#).
- Editorial changes to [§ 1 Motivation](#) and [§ 6.1 Tag Type](#) to support our design choice.
- Added the reference to [\[LIBC++\]](#) in [§ 3 Implementation](#).

§ 10.3. R1 → R2

Applied feedback from [\[SAN\]](#) Meeting reviews.

- Reverted design to "option A" proposed in [\[P1072R0\]](#).
- Switched from `resize_uninitialized` to `resize_default_init`.
- Added discussion of alternatives considered.
- Specified allocator interaction.
- Added wording.

§ 10.4. R0 → R1

Applied feedback from LEWG [\[post-Rapperswil\]](#) Email Review:

- Shifted to a new vocabulary types: `storage_buffer` / `storage_node`
 - Added presentation of `storage_buffer` as a new container type
 - Added presentation of `storage_node` as a node-like type
- Added discussion of design and implementation considerations.
- Most of [\[P1010R1\]](#) Was merged into this paper.

§ 11. Acknowledgments

Thanks go to **Arthur O’Dwyer** for help with wording and proof reading, to **Jonathan Wakely** for hunting down the language that makes `basic_string` allocator-aware, and to **Glen Fernandes**, **Corentin Jabot**, **Billy O’Neal**, and **Mathias Stearn** for design discussion. Special thanks to **Eric Fiselier** for providing the implementation.

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